QuoteSpace — Entity Relationship Diagram (ERD)

System Database Architecture & Schema Specification

1. Entity Relationship Overview

The QuoteSpace database consists of two primary entities: **User** and **Quote**. There exists a **One-to-Many (1:N)** relationship where one registered user can author multiple quotes, while each quote belongs strictly to exactly one registered user.

Primary Entity	Cardinality	Related Entity	Foreign Key	Deletion Behavior
User (users)	**1 : N** (One-to-Many)	**Quote** (quotes)	user_id -> users.id	CASCADE DELETE

2. Detailed Entity Schema Specification

Entity: Users (Table: users)

Column Name	Data Type	Key / Index	Constraints / Nullable	Description
id	BIGINT UNSIGNED	**PK** (Primary Key)	AUTO_INCREMENT, NOT NULL	Unique user identifier
name	VARCHAR(255)	-	NOT NULL (Length: 2-50)	User full display name
email	VARCHAR(255)	**UNIQUE INDEX**	NOT NULL, Valid Email	Authentication email address
password	VARCHAR(255)	-	NOT NULL (Bcrypt Hashed)	Securely hashed password
remember_token	VARCHAR(100)	-	NULLABLE	Persistent session token
created_at	TIMESTAMP	-	NULLABLE	Record creation timestamp
updated_at	TIMESTAMP	-	NULLABLE	Record update timestamp

Entity: Quotes (Table: quotes)

Column Name	Data Type	Key / Index	Constraints / Nullable	Description
id	BIGINT UNSIGNED	**PK** (Primary Key)	AUTO_INCREMENT, NOT NULL	Unique quote identifier
user_id	BIGINT UNSIGNED	**FK, INDEX**	NOT NULL, ON DELETE CASCADE	References users(id)
content	VARCHAR(255)	**FULLTEXT / INDEX**	NOT NULL (Length: 3-155)	Quote text content
created_at	TIMESTAMP	**INDEX (created_at DESC)**	NULLABLE	Publication timestamp
updated_at	TIMESTAMP	-	NULLABLE	Last edit timestamp

3. Mermaid ER Diagram Source Representation

```
erDiagram
USERS ||--o{ QUOTES : "authors (1:N)"
USERS {
bigint id PK
string name "2-50 chars"
string email UK "unique"
string password "hashed"
timestamp created_at
timestamp updated_at
}
QUOTES {
bigint id PK
bigint user_id FK "references users(id)"
string content "3-155 chars"
timestamp created_at "indexed"
timestamp updated_at
}
```